Accepted Manuscript



A physiologically based pharmacokinetic model for valacyclovir established based on absolute expression quantity of hPEPT1 and its application

Le Sun, Chao Wang, Youxi Zhang

PII: S0928-0987(18)30360-9
DOI: doi:10.1016/j.ejps.2018.07.057
Reference: PHASCI 4633

To appear in: *European Journal of Pharmaceutical Sciences*

Received date: 15 April 2018
Revised date: 5 July 2018
Accepted date: 30 July 2018

Please cite this article as: Le Sun, Chao Wang, Youxi Zhang , A physiologically based pharmacokinetic model for valacyclovir established based on absolute expression quantity of hPEPT1 and its application. Phasci (2018), doi:10.1016/j.ejps.2018.07.057

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

ACCEPTED MANUSCRIPT

Article

Title:

A Physiologically Based Pharmacokinetic Model for Valacyclovir Established Based on Absolute Expression Quantity of hPEPT1 and Its Application

Authors:

Le Sun [a,*], Chao Wang [b], Youxi Zhang [c]

a. Department of Pharmaceutics, School of Pharmacy, China Medical University, No. 77 Puhe Road, Shenyang North New Area, Shenyang 110122, China

b. Liaoning Institute for Drug Control, No. 7 Chongshan West Road, Huanggu Area, Shenyang 110122, China

c. Department of Pharmacy, The Fourth Affiliated Hospital, China Medical University, No. 4 Chongshan East Road, Huanggu Area, Shenyang 110122, China

*Corresponding author

Dr. Le Sun Tel: +86-18909833928

Email address: lesun@cmu.edu.cn

ACCEPTED MANUSCRIPT

Abstract: In this study, a physiologically based pharmacokinetic (PBPK) model was established for valacyclovir based on absolute expression quantity of hPEPT1 along the entire length of the human intestine and other reliable *in vitro, in vivo* observed data. The PBPK model-3 defined acyclovir as metabolite of valacyclovir and simulated the plasma concentration-time profiles of valacyclovir and acyclovir simultaneously. It was validated strictly by a series of observed plasma concentration-time profiles. The average fold error (AFE) and absolute average fold error (AAFE) values were all smaller than 2. Then, it was used to quantitatively evaluate the effect of hPEPT1, luminal degradation rate, drug release rate and gastric residence time on the oral absorption of valacyclovir and acyclovir. The PBPK model-3 suggests that mainly 75% of valacyclovir was absorbed by active transport of hPEPT1. The luminal degradation of valacyclovir in the upper intestinal lumen cannot be considered the only reason for its incomplete bioavailability. The plasma concentration-time profiles of valacyclovir and its metabolite acyclovir were not sensitive to dissolution rate faster than $T_{85\%}$=120 min. Prolonged gastric residence time of sustained release tablet can improve the oral absorption of valacyclovir. All in all, the PBPK model-3 in this study is reliable and accurate. It is useful for the research of clinical application and dosage forms design of valacyclovir.

Chemical compounds studied in this article: Valacyclovir (PubChem CID: 60773); acyclovir (PubChem CID: 2022)

ACCEPTED MANUSCRIPT

Keyword: PBPK model, absolute expression quantity of transporter, valacyclovir, *in vivo* pharmacokinetics, oral absorption, hPEPT1

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

1. Introduction

Valacyclovir is the L-valyl ester prodrug of acyclovir prescribed for the treatment of viral infections (Shinkai and Ohta, 1996). Following its oral absorption, valacyclovir is rapidly and virtually converted to the active drug acyclovir by hydrolases which is a key step for the therapeutic efficacy (de Miranda and Burnette, 1994; Lai et al., 2008). In fact, the prodrug was undetectable in plasma 3 hours after the dose, whereas the plasma acyclovir exposure was similar to that achieved after intravenous doses of acyclovir (Soul-Lawton et al., 1995).

Although valacyclovir has been demonstrated to be chemically stable in acidic pH, about 16% of the human dose is enzymatically hydrolyzed in human intestinal lumen during the small intestinal transit (~ 3h) contributing to reduce the drug oral bioavailability (Granero and Amidon, 2006). Conversely, valacyclovir is a model of hPEPT1-targeting and previous studies revealed that the C_{max} and area under the curve (*AUC*) of acyclovir were about up to 6- and 3-fold lower, respectively, in PepT1 knockout mice than in wild-type animals (Yang et al., 2013). Similarly, the intestinal permeability of the prodrug was about 10% of that in wild-type animals (Yang and Smith, 2013). In this sense, sustained-released formulations able to increase the mean residence time in the upper part of gastrointestinal tract and have been suggested to improve the disposition (Upadhyay et al., 2014).

Quantitative *in vivo* investigation regarding the effect of the above factors on the oral absorption of valacyclovir involves ethical and economic issues as well as a backbreaking work. On the other hand, *in silico* models simulating the drug

ACCEPTED MANUSCRIPT

absorption, distribution, metabolism and excretion (ADME) in human is a helpful tool in drug development. Previous models using advanced compartmental absorption and transit model (ACAT) simulated the adsorption and pharmacokinetics of valacyclovir in human. However, the pre-systemic hydrolysis, mean residence time, and the expression levels of hPEPT1 were not considered (Bolger et al., 2009). Altogether, these parameters are critical to the accuracy and reliability of the *in silico* model (Abuasal et al., 2012; Al Feteisi et al., 2015; Harwood et al., 2016a; Harwood et al., 2016b). Furthermore, the modeling of the *in vivo* pharmacokinetic profiles of acyclovir following escalating oral doses of valacyclovir in wildtype and PepT1 knockout mice was successfully established (Yang and Smith, 2017). However, it would be more useful to translate such information to a clinical scenario instead of animals.

Therefore, in this study, a physiologically based pharmacokinetic (PBPK) model was established to simulate the plasma concentration-time profile in human after oral administration of valacyclovir considering the absolute expression levels of hPEPT1 along the entire length of the human intestine, the gastric residence time, as well as the hydrolyzing, the drug release and the transporter kinetic parameters.

2. Methods

2.1 Human Gastrointestinal Physiology

The default human fasted physiological model provided in GastroPlus™ (version 9.5, Simulation Plus, Lancaster, CA, USA) including the absorption scale factors

ACCEPTED MANUSCRIPT

(ASF) obtained by Opt LogD Model SA/V6.1 used in this study were shown in Table 1.

2.2 Absolute Content of hPEPT1 in the Intestinal Segment

The absolute expression of the transporter in duodenum (2574 ± 553 fmol/mg), jejunum (3720 ± 461 fmol/mg) and ileum (4915 ± 1383 fmol/mg) (Drozdzik et al., 2014; Groer et al., 2013) were converted to mg/intestinal segment considering the molecular weight of the transporter (78.8 Kdalton) (Merlin et al., 2001), the density of the tissue (1×10^3 mg/mL), and the volume of segment intestinal wall ($\nu_{(i)}$). The $\nu_{(i)}$ was estimated by the difference between the total volume of the intestinal segment ($\nu_{(total)}$) and the corresponding default volumes of the lumen ($\nu_{(lumen)}$) (Table 1). Finally, $\nu_{(total)}$ was calculated as $\nu_{(total)} = \pi(r_{(i)}+t)^2 \times l_{(i)}$, where $r_{(i)}$ is the radius (cm) of the intestinal segment obtained directly from GastroPlus™ (Table 1), t is the intestinal thickness stated as 0.2 cm (Peng L., 2016), and $l_{(i)}$ is the length of the intestinal segment obtained directly from GastroPlus™ (Table 1). For the caecum and colon, the transport expressions were adjusted to zero (Merlin et al., 2001).

2.3 The Transporter Kinetic Parameters (K_m and V_{max}) between Valacyclovir and hPEPT1

The values for K_m and V_{max} corresponding to the hPEPT1-mediated transport of valacyclovir inputted into the model were 1.5 mg/L (4.6 μM) and 0.45 μg/s/mg-hPEPT1, respectively (Bhardwaj et al., 2005). The V_{max} was optimized when

ACCEPTED MANUSCRIPT

establishing the model to 5.34-fold of the mean value of medium and high expression transfected MDCK cell (15.60 nanomoes/mg-hPEPT1).

2.4 Valacyclovir Luminal Degradation

The half-life ($t_{1/2}$) of vacyclovir in phosphate buffer at pH values of 4.11 (39.6 h), 4.92 (20.2 h), 6.32 (15.54 h), 7.39 (10.06 h), and 8.09 (7.48 h) were divided by 1.66 to estimate the half-life vs pH curve in human GI tract. This was the ratio of the $t_{1/2}$ in phosphate buffer pH 6.32 and that from human intestinal fluid (9.36) at similar pH (Granero and Amidon, 2006).

2.5 The Passive Human Intestinal Membrane Permeability of Valacyclovir

After oral administration of valacyclovir, the prodrug and the acyclovir are simultaneously present in the intestinal lumen. Based on the molecular weight and log*P* values the acyclovir presents higher passive permeability (P_{app}) compared to the parent compound (Sun et al., 2013). Thus, the passive permeability of acyclovir was used in PBPK model instead of the passive permeability of valacyclovir. The passive permeability of acyclovir was 0.25 $\times 10^{-4}$cm/s predicted by ADMET Predictor™ in GastroPlus™. Then the value was optimized to 0.19 $\times 10^{-4}$cm/s by PBPK model simulated plasma concentration-time profile of oral immediate release acyclovir tablet.

2.6 The Hydrolysis of Valacyclovir in Enterocyte, Liver and Blood

ACCEPTED MANUSCRIPT

The K_m value corresponding the hydrolysis of valacyclovir by liver, enterocyte and blood was set to 61.62 mg/L (190 μM) (Kim et al., 2003). The V_{max} value corresponding the hydrolysis of valacyclovir by liver, enterocyte and blood was set to 0.09 mg/s/mg-enz. It is optimized based on data in reference (Kim et al., 2003). PBPK model in GastroPlus version 9.5 does not allow for metabolic conversion directly in the blood or plasma and lung was selected as a surrogate tissue to account for this conversion.

The expression of hydrolase in liver and lung were 0.12 and 0.001 mg/g-tissue, respectively. The order of magnitude of them was the same with other enzymes in the corresponding tissue. The expression of hydrolase along the GI tract was shown in Table 1.

2.7 The establishment of PBPK Models

The PBPK model-1 and model-2 simulating the plasma concentration-time profiles after intravenous injection of acyclovir solution and oral administration of acyclovir tablet were established, respectively. The PBPK modle-3 was established to simultaneously simulate the plasma concentration-time profiles of valacyclovir and its metabolite acyclovir. The luminal degradation of valacyclovir, the active transport of valacyclovir, the hydrolysis of valacyclovir in liver, enterocyte and blood were all considered in PBPK modle-3.

The basic physicochemical and biopharmaceutical properties inputted in PBPK models were listed in Table 2. The parameters in PBPK model-1 and model-2 were all the same except dosage and formulation option. The pharmacokinetic parameters of

ACCEPTED MANUSCRIPT

acyclovir in the PBPK model-2 were used to simulate the plasma concentration-time profile of acyclovir in PBPK model-3. In addition, the demographic information of PBPK models was chose based on the observed study in literature (Soul-Lawton et al., 1995, Neves et al., 2010) (Table 3). The tissue: plasma partition coefficients (Kp) of lung, spleen, liver, gut, adipose, muscle, heart, brain, kidney, skin, reproductive organs, red bone marrow, yellow bone marrow and rest-of-body was automatic calculated by GastroPlus™ based on Poulin method.

2.8 Parameter Sensitivity Analysis

In order to find out the key factors influenced the simulated plasma concentrations of valacyclovir and acyclovir, parameter sensitivity analysis (PSA) was performed using PBPK model-3. The initial input values were varied by multiplying them with 10 scaling factors in the range of 0.1-10 to allow an order of magnitude increase or decrease. C_{max} and $AUC_{0-\infty}$ of valacyclovir and acyclovir were then evaluated at each of the 10 scaling factors for each of the parameters studied.

2.9 Model Validation

The PBPK model-1 and 2 were validated by modeling the observed plasma concentration-time profiles of acyclovir after its intravenous and oral administration from previous studies (Soul-Lawton et al., 1995; Kubbinga et al., 2015), whereas for the model-3 the validation was accomplished by simultaneous fitting of valacyclovir and acyclovir plasma concentration-time profiles following the oral administration of the prodrug (Neves et al., 2010). In addition, the PBPK model-3 was used to simulate

ACCEPTED MANUSCRIPT

the plasma concentration-time profiles following oral administration of 250, 500, 1000, 1500 and 2000 mg of valacyclovir, respectively. Then the simulated plasma concentration-time profiles were validated by observed data (Weller et al., 1993).

The average fold error (AFE) and absolute average fold error (AAFE) was calculated as below:

$$\mathrm{AFE} = 10^{\frac{1}{n}*\sum \log(\frac{\mathrm{predicted}_i}{\mathrm{observed}_i})} \quad (1)$$

$$\mathrm{AAFE} = 10^{\frac{1}{n}*\sum \left|\log(\frac{\mathrm{predicted}_i}{\mathrm{observed}_i})\right|} \quad (2)$$

Where n is the number of time points at which the concentration was determined. $\mathrm{Predicted}_i$ and $\mathrm{observed}_i$ are the predicted and observed concentrations at a given time point *i*. The AFE indicates whether the predicted profile underestimate or overestimates the observed values. The AAFE quantifies the absolute error from the true values (as shown in Eq. 2). The AFE and AAFE of <2 can be considered as a successful simulation.

2.10 Model Application

Once the predictive performance of the PBPK model-3 was fully validated, it was used to quantitative evaluate the effect of hPEPT1 on the oral absorption of valacyclovir. The fraction of valacyclovir dose absorbed was simulated considering the presence and absence of hPEPT1 expression in the intestine, respectively.

The simulated fraction of valacyclovir dose absorbed at each segment of intestine under both with hPEPT1 and without hPEPT1 conditions were compared with the experimental value obtained by wild-type mice, PepT1 knockout mice and

ACCEPTED MANUSCRIPT

wild-type rat. Since the active hPEPT1-mediated transport is by far the most important mechanism of intestinal absorption of valacyclovir, the predicted active transport was used to compare with experimental data in mice and rats. In addition, the predicted active transport calculated based on the absolute expression quantity of hPEPT1. Thus, these data were directly compared with effective permeability at corresponding segment of intestine in wild type mice (Yang and Smith, 2013) and wild type rat. Besides, the passive effective permeability of valacyclovir used in PBPK model-3 was compared with the experimental effective permeability in PepT1 knockout type mice (Yang and Smith, 2013).

The PBPK model-3 was also used to quantitative evaluate the effect of luminal degradation on the oral absorption of valacyclovir. Based on the *in vitro* luminal degradation half-life *vs* pH curve in human GI tract (Granero and Amidon, 2006) , the slower and quicker luminal degradation rates were respectively inputted into the PBPK model-3, and used to simulate the plasma concentration-time profiles of valacyclovir and acyclovir.

In order to quantitative evaluate the effect of dissolution rate on the oral absorption of valacyclovir, the simulated plasma concentration-time profiles of valacyclovir and acyclovir were obtained, respectively, at different dissolution rates ($T_{85\%}$=15, 30, 45, 60, 90, 120, 180, 240 and 300 min) by the validated PBPK model-3. The simulated plasma concentration-time profile of $T_{85\%}$=15 min [＞85% solubility (pH 1.2-6.8) in 15 min; “very rapid dissolution”] as the reference, were compared with those of other dissolution rates.

ACCEPTED MANUSCRIPT

The gastric residence dosage form with 100% drug release within 12 h was chosen, and the plasma concentration-time profiles of valacyclovir and acyclovir were simulated by PBPK model-3 setting 1, 2, 3, 4, 5, 6, 7 and 8 h of gastric residence time, respectively. In addition, sustained release tablet without prolonged gastric residence was also simulated by PBPK model-3.

3. Results

3.1 Validation of the PBPK Models

The predicted and observed concentrations of valacyclovir and acyclovir obtained from the PBPK models are present in Figure 1, whereas the simulated profiles of acyclovir are presented on Figure 2. The predicted and observed values of C_{max} and $AUC_{0-\infty}$ after oral administration of valacyclovir to normal volunteers are shown in the Table 4. As shown in Figure 1 and 2, the observed and predicted plasma concentration-time profiles were superimposed. In all cases the AFE and AAFE values for model-1, model-2, and model-3 were less than 2 indicating an adequate fitting. These results indicate that the PBPK model-1, model-2 and model-3 are accurate and reliable.

3.2 Parameter Sensitivity Analysis

As shown in Figure 3, PSA suggested that the predicted C_{max} and $AUC_{0-\infty}$ of valacyclovir and acyclovir were most sensitive to changes in clearance (CL) in liver and insensitive to CL in blood. C_{max} and $AUC_{0-\infty}$ of valacyclovir and acyclovir were

ACCEPTED MANUSCRIPT

also insensitive to solubility, particle radius and passive permeability.

As shown in Figure 4, PSA suggested that the predicted C_{max} and $AUC_{0-\infty}$ of valacyclovir and acyclovir were most sensitive to changes in V_{max} of hPEPT1, K_m and V_{max} of hydrolase in liver. However, they were insensitive to K_m and V_{max} of hydrolase in blood and enterocyte. The predicted $AUC_{0-\infty}$ of valacyclovir and acyclovir were not increased when the scaling factor of hPEPT1's V_{max} was larger than 1. The expression of hPEPT1 in duodenum has small effect on the oral absorption of valacyclovir.

3.3 Effect of hPEPT1 on the Oral Absorption of Valacyclovir

The simulated fraction of valacyclovir dose absorbed when considering the presence and absence of hPEPT1 intestinal expression are shown in Figure 5. The fraction of valacyclovir dose absorbed increased from 25.9% to 93.7% when the simulations were performed with hPEPT1. Mainly, 75% of valacyclovir was absorbed by active transport of hPEPT1. The fraction of valacyclovir dose absorbed in the absence of intestinal hPEPT1-mediated transport (25.9%) was close to the oral bioavailability of acyclovir in human (Fletcher and Bean, 1985). This further proofs the accuracy and reliability of PBPK model-3. The duodenum and jejunum accounted for 9% and 64% of the dose absorbed. It indicated that valacyclovir was absorbed mainly in the upper segment of small intestine.

The mice, rats and humans all have large absorption at duodenum, jejunum and ileum. However, their absorptions at colon were all small (Figure 6(a)). The effective permeabilities across the jejunum and ileum in wild type rats were measured in our laboratory. The passive permeability of valacyclovir in PBPK model-3 was compared

ACCEPTED MANUSCRIPT

with effective permeability of PepT1 knockout mice (Figure 6(b)). They were both much smaller than the absorption through hPEPT1.

3.4 Effect of Luminal Degradation on the Oral Absorption of Valacyclovir

The predicted plasma concentration-time profiles of valacyclovir and acyclovir under different luminal degradation rate are shown in the Figure 7. Compared with the simulation using the observed luminal degradation rate, the relative errors of C_{max} and $AUC_{0-\infty}$ were all within 20% no matter when there was no luminal degradation or 4 times of luminal degradation rate.

3.5 Effect of Drug Release Rate on the Oral Absorption of Valacyclovir

The predicted plasma concentration-time profiles of valacyclovir and acyclovir under different dissolution rates are shown in the Figure 8. The relative error of C_{max} and $AUC_{0-\infty}$ were within 20% up to $T_{85\%}$=120 min. This indicated that the plasma concentration-time profiles of valacyclovir and its metabolite acyclovir were not sensitive to dissolution rate. However, when the dissolution rate was slower than $T_{85\%}$=120 min, the C_{max} and $AUC_{0-\infty}$of valacyclovir and acyclovir decreased quickly.

3.6 Effect of Gastric Residence Time on the Oral Absorption of Valacyclovir Sustained Release Tablet

The predicted plasma concentration-time profiles of valacyclovir and acyclovir under different gastric residence time were shown in the Figure 9. For sustained release dosage form without prolonged gastric residence, the C_{max} and $AUC_{0-\infty}$ of valacyclovir and acyclovir was small. The C_{max} and $AUC_{0-\infty}$ of valacyclovir and

ACCEPTED MANUSCRIPT

acyclovir increased with the prolonging gastric residence time (gastric residence time=1, 2, 3, 4, 5, 6, 7 and 8). This indicated that prolonged the gastric residence time of sustained release tablet can increase the C_{max} and $AUC_{0-\infty}$ of valacyclovir and acyclovir. The C_{max}, $AUC_{0-\infty}$ and T_{max} of acyclovir after oral administration of valacyclovir sustained release tablet ($T_{100\%}$=12 h) with gastric residence time 8 h were 1498.18 ng/mL, 13479.9 ng·h/mL and 8.533h, respectively. The C_{max}, $AUC_{0-\infty}$ and T_{max} of acyclovir after oral administration of valacyclovir immediate release tablet were 4555.31 ng/mL, 14271.9 ng·h/mL and 1.86667 h, respectively. The C_{max} of gastric residence sustained release tablet is about 3 times smaller than the immediate release tablet. However, the $AUC_{0-\infty}$ of them was almost the same.

4. Discussion

Bolger et al. (Bolger et al., 2009) and Smith et al. (Yang and Smith, 2017) simulated the *in vivo* pharmacokinetic profiles of acyclovir following oral administration of valacyclovir in humans and mice using ACAT model implemented in GastroPlus™. In this study, the acyclovir was defined as the metabolite of valacyclovir in PBPK model-3. The plasma concentration-time profiles of valacyclovir and acyclovir were simultaneously simulated and were both validated by observed plasma concentration-time profiles.

The PBPK model-3 was established adding the hydrolysis of valacyclovir in liver, enterocyte and blood. Valacyclovir is rapidly and nearly completely hydrolyzed to acyclovir by hepatic and intestinal first-pass metabolism (de Miranda and Burnette,

ACCEPTED MANUSCRIPT

1994; Lai et al., 2008). C_{max} and $AUC_{0-\infty}$ of valacyclovir and acyclovir were sensitive to change in K_m and V_{max} of hydrolase in liver and insensitive to change in K_m and V_{max} of hydrolase in blood. However, we consider that it is necessary to add the hydrolase in blood. This is because that the predicted concentration of valacyclovir in plasma did not almost decrease to 0 within 6 hours just as the observed value even though the V_{max} of hydrolase in liver was set largely enough in the process of establishing PBPK model-3. In addition, *in vitro* studies showed 15% of valganciclovir was degraded over 60 minutes in whole mouse blood (Lukacova et al., 2016). Thus, the degradation of valacyclovir in blood was considered in PBPK model-3.

Since valacyclovir is quite soluble in water (174 mg/mL) (Beauchamp et al., 1992) its C_{max} and $AUC_{0-\infty}$ values were insensitive to solubility and particle radius. Valacyclovir is mainly absorbed through abundant hPEPT1 expressed along the small intestine. Thus, C_{max} and $AUC_{0-\infty}$ of valacyclovir and acyclovir were also insensitive to the passive permeability and were minor affected by the expression of hPEPT1 in duodenum.

The C_{max} and $AUC_{0-\infty}$ of valayclovir decreased 5.06% and 6.37% compared with in the absence of luminal degradation simulating by PBPK model-3. They are not decreased too much. This is because that valacyclovir has a relatively slower luminal degradation rate at the upper segment of intestine where the pH is relatively lower and valayclovir was absorbed quickly. This result is consisted with the conclusion of Mizuma et al. (Mizuma, 2010). Thus, the luminal degradation of valacyclovir in the

ACCEPTED MANUSCRIPT

upper intestinal lumen cannot be considered the only reason for its incomplete bioavailability (54.2%) (Soul-Lawton et al., 1995)

The C_{max} and $AUC_{0-\infty}$ of valacyclovir and acyclovir were insensitive to the dissolution rate up to $T_{85\%}$=120 min, which can be justified by the high solubility of the prodrug. However, at longer dissolution rates (> $T_{85\%}$=120 min) more drug would be released at the lower segment of small intestine where the pH is high, drug degradation rate is quick, and the drug absorption rate is slow. As consequence, the oral absorption of valacyclovir would decrease rapidly.

On the other hand, sustained release tablets with gastroretentive properties could be an alternative since more drug would be released at the proximal intestine where absorption is rapid, and metabolism is slow. Kagan L et al. (Kagan and Hoffman, 2008) concluded the same trend about the C_{max}, $AUC_{0-\infty}$ and T_{max} of acyclovir following the administration of valacyclovir as oral bolus by gastric infusion over 4 h or 8 h to rats. Therefore, the PBPK model-3 can be a valuable tool during the development of gastroretentive sustained release tablets.

Yet, the absolute expression of hPEPT1 along the intestine and the *in vitro-in vivo* extrapolation scaling factors for transporters were critical for the accuracy of the PBPK model (Harwood et al., 2016a; Harwood et al., 2016b). However, the lack of accurate methods such as liquid chromatography-tandem mass spectrometry to detect the absolute expression quantity of intestinal uptake and efflux transporter proteins and controversial data from different laboratories can be considered a limitation for establishing such PBPK model.

ACCEPTED MANUSCRIPT

5. Conclusions

In the present study, the PBPK model-3 simulating the plasma concentration-time profiles of valacyclovir and acyclovir was established successfully. The method was based on reliable *in vitro*/*in vivo* data and was validated strictly by a series of observed plasma concentration-time profiles. The PBPK model-3 suggests that mainly 75% of valacyclovir is absorbed by active transport of hPEPT1. The luminal degradation of valacyclovir in the upper intestinal lumen cannot be considered the only reason for its incomplete bioavailability. The plasma concentration-time profiles of valacyclovir and its metabolite acyclovir were not sensitive to dissolution rate faster than $T_{85\%}$=120 min. Prolonged gastric residence time of sustained release tablet can improve the oral absorption of valacyclovir. All in all, the PBPK model-3 in this study is reliable and accurate. It is useful for the research of clinical application and dosage forms design of valacyclovir.

Acknowledgments

This work was supported by National Natural Science Foundation of China (No. 81603055) and National Natural Science Foundation of China (No. 81703447).

ACCEPTED MANUSCRIPT

Legend to Figures

Figure 1. Predicted and observed plasma concentration-time profiles of valacyclovir and acyclovir: (a) after intravenous infusion of acyclovir (350 mg) over 1 h (n=12); (b) after oral administration of acyclovir (200 mg) (n=12); (c) after administration of valacyclovir (1000 mg) (n=46). The observed data were presented as mean±SD.

Figure 2. The predicted and observed (n=8) plasma concentration-time profiles of acyclovir after oral administration of 250, 500, 1000, 1500 and 2000 mg valacyclovir, respectively.

Figure 3. Sensitivity of predicted C_{max} (**a**) and $AUC_{0-\infty}$ (**b**) of valacyclovir and predicted C_{max} (**c**) and $AUC_{0-\infty}$ (**d**) of acyclovir using PBPK model-3. Parameters were changed by multiplying the initial input values with scaling factors in the range of 0.1-10.

Figure 4. Sensitivity of predicted C_{max} (**a**) and $AUC_{0-\infty}$ (**b**) of valacyclovir and predicted C_{max} (**c**) and $AUC_{0-\infty}$ (**d**) of acyclovir using PBPK model-3. Parameters were changed by multiplying the initial input values with scaling factors in the range of 0.1-10.

Figure 5. Contribution of specific intestinal regions in the oral absorption of 1000 mg

ACCEPTED MANUSCRIPT

valacyclovir in human with hPEPT1 and without hPEPT1.

Figure 6. Absolute expression quantity of hPEPT1 in human duodenum, jejunum, ileum and colon used in PBPK model-3, as well as effective permeability of valacyclovir in wild type mice duodenum, jejunum, ileum and colon (mean±S.E., n=4-5) and wild type rat jejunum and ileum (mean±SD, n=3) (**a**); The passive permeability of valacyclovir used in PBPK model-3 and effective permeability of valacyclovir in PepT1 knockout mice duodenum, jejunum, ileum and colon (mean±S.E., n=4-5) (**b**).

Figure 7. The effect of luminal degradation on the predicted plasma concentration-time profiles of valacyclovir (**a**) and acyclovir (**b**) after oral administration of 1000 mg valacyclovir.

Figure 8. The effect of release rate on the predicted plasma concentration-time profiles of valacyclovir (**a**) and acyclovir (**b**) after oral administration of 1000 mg valacyclovir.

Figure 9. The effect of gastric residence time on the predicted plasma concentration-time profiles of valacyclovir (**a**) and acyclovir (**b**) after oral administration of valacyclovir sustained release tablet.

ACCEPTED MANUSCRIPT

Reference:

Abuasal, B.S., Bolger, M.B., Walker, D.K., Kaddoumi, A., 2012. In silico modeling for the nonlinear absorption kinetics of UK-343,664: a P-gp and CYP3A4 substrate. Mol Pharm 9, 492-504.

Al Feteisi, H., Achour, B., Barber, J., Rostami-Hodjegan, A., 2015. Choice of LC-MS methods for the absolute quantification of drug-metabolizing enzymes and transporters in human tissue: a comparative cost analysis. AAPS J 17, 438-446.

Beauchamp, L. M., Orr, G. F., Miranda, P. D., Bumette, T. Krenitsky, T. A. Amino acid este prodrugs of acyclovir. Antiviral Chemistry and Chemotherapy 3, 157-164.

Bhardwaj, R.K., Herrera-Ruiz, D., Sinko, P.J., Gudmundsson, O.S., Knipp, G., 2005. Delineation of human peptide transporter 1 (hPepT1)-mediated uptake and transport of substrates with varying transporter affinities utilizing stably transfected hPepT1/Madin-Darby canine kidney clones and Caco-2 cells. J Pharmacol Exp Ther 314, 1093-1100.

Bolger, M.B., Lukacova, V., Woltosz, W.S., 2009. Simulations of the nonlinear dose dependence for substrates of influx and efflux transporters in the human intestine. AAPS J 11, 353-363.

de Miranda, P., Burnette, T.C., 1994. Metabolic fate and pharmacokinetics of the acyclovir prodrug valaciclovir in cynomolgus monkeys. Drug Metab Dispos 22, 55-59.

Drozdzik, M., Groer, C., Penski, J., Lapczuk, J., Ostrowski, M., Lai, Y., Prasad, B., Unadkat, J.D., Siegmund, W., Oswald, S., 2014. Protein abundance of clinically relevant multidrug transporters along the entire length of the human intestine. Mol Pharm 11, 3547-3555.

Fletcher, C., Bean, B., 1985. Evaluation of oral acyclovir therapy. Drug Intell Clin Pharm 19, 518-524.

Granero, G.E., Amidon, G.L., 2006. Stability of valacyclovir: implications for its oral bioavailability. Int J Pharm 317, 14-18.

ACCEPTED MANUSCRIPT

Groer, C., Bruck, S., Lai, Y., Paulick, A., Busemann, A., Heidecke, C.D., Siegmund, W., Oswald, S., 2013. LC-MS/MS-based quantification of clinically relevant intestinal uptake and efflux transporter proteins. J Pharm Biomed Anal 85, 253-261.

Harwood, M.D., Achour, B., Neuhoff, S., Russell, M.R., Carlson, G., Warhurst, G., Amin, R.-H., 2016a. In Vitro-In Vivo Extrapolation Scaling Factors for Intestinal P-Glycoprotein and Breast Cancer Resistance Protein: Part I: A Cross-Laboratory Comparison of Transporter-Protein Abundances and Relative Expression Factors in Human Intestine and Caco-2 Cells. Drug Metab Dispos 44, 297-307.

Harwood, M.D., Achour, B., Neuhoff, S., Russell, M.R., Carlson, G., Warhurst, G., Rostami-Hodjegan, A., 2016b. In Vitro-In Vivo Extrapolation Scaling Factors for Intestinal P-glycoprotein and Breast Cancer Resistance Protein: Part II. The Impact of Cross-Laboratory Variations of Intestinal Transporter Relative Expression Factors on Predicted Drug Disposition. Drug Metab Dispos 44, 476-480.

Kagan, L., Hoffman, A., 2008. Selection of drug candidates for gastroretentive dosage forms: pharmacokinetics following continuous intragastric mode of administration in a rat model. Eur J Pharm Biopharm 69, 238-246.

Kim, I., Chu, X.Y., Kim, S., Provoda, C.J., Lee, K.D., Amidon, G.L., 2003. Identification of a human valacyclovirase: biphenyl hydrolase-like protein as valacyclovir hydrolase. J Biol Chem 278, 25348-25356.

Kubbinga, M., Nguyen, M.A., Staubach, P., Teerenstra, S., Langguth, P., 2015. The Influence of Chitosan on the Oral Bioavailability of Acyclovir--a Comparative Bioavailability Study in Humans. Pharm Res 32, 2241-2249.

ACCEPTED MANUSCRIPT

Lai, L., Xu, Z., Zhou, J., Lee, K.D., Amidon, G.L., 2008. Molecular basis of prodrug activation by human valacyclovirase, an alpha-amino acid ester hydrolase. J Biol Chem 283, 9318-9327.

Lukacova, V., Goelzer, P., Reddy, M., Greig, G., Reigner, B., Parrott, N., 2016. A Physiologically Based Pharmacokinetic Model for Ganciclovir and Its Prodrug Valganciclovir in Adults and Children. AAPS J 18, 1453-1463.

Merlin, D., Si-Tahar, M., Sitaraman, S.V., Eastburn, K., Williams, I., Liu, X., Hediger, M.A., Madara, J.L., 2001. Colonic epithelial hPepT1 expression occurs in inflammatory bowel disease: tansport of bacterial peptides influences expression of MHC Class 1 molecules. GASTROENTEROLOGY 120, 1666-1679.

Mizuma, T., 2010. Kinetic assessment of luminal degradation of orally effective prodrugs for rational drug development. J Pharm Sci 99, 1078-1086.

Neves, R., Almeida, S., Filipe, A., Spinola, A.C., Abolfathi, Z., Levesque, A., Ortuno, J., Torns, A., 2010. Bioequivalence studies of two different film-coated tablet formulations of valacyclovir of two different strengths in healthy volunteers. Arzneimittelforschung 60, 273-281.

Peng L., L.H., Xu F. Y., Han Y. X., Gu L. F., Ye C. Y., 2016. Multi-temporal and multiangle analyses on MSCTE of each segment of the small intestine in normal adults. J Med Imaging 26, 1444-1448.

Shinkai, I., Ohta, Y., 1996. New drugs--reports of new drugs recently approved by the FDA. Valacyclovir. Bioorg Med Chem 4, 1-2.

Soul-Lawton, J., Seaber, E., On, N., Wootton, R., Rolan, P., Posner, J., 1995. Absolute bioavailability and metabolic disposition of valaciclovir, the L-valyl ester of acyclovir, following oral administration to humans. Antimicrob Agents Chemother 39, 2759-2764.

Sun, L., Liu, X., Xiang, R., Wu, C., Wang, Y., Sun, Y., Sun, J., He, Z., 2013. Structure-based prediction

ACCEPTED MANUSCRIPT

of human intestinal membrane permeability for rapid in silico BCS classification. Biopharm Drug Dispos 34, 321-335.

Upadhyay, P., Nayak, K., Patel, K., Patel, J., Shah, S., Deshpande, J., 2014. Formulation development, optimization, and evaluation of sustained release tablet of valacyclovir hydrochloride by combined approach of floating and swelling for better gastric retention. Drug Deliv Transl Res 4, 452-464.

Weller, S., Blum, M.R., Doucette, M., Burnette, T., Cederberg, D.M., de Miranda, P., Smiley, M.L., 1993. Pharmacokinetics of the acyclovir pro-drug valaciclovir after escalating single- and multiple-dose administration to normal volunteers. Clin Pharmacol Ther 54, 595-605.

Yang, B., Hu, Y., Smith, D.E., 2013. Impact of peptide transporter 1 on the intestinal absorption and pharmacokinetics of valacyclovir after oral dose escalation in wild-type and PepT1 knockout mice. Drug Metab Dispos 41, 1867-1874.

Yang, B., Smith, D.E., 2013. Significance of peptide transporter 1 in the intestinal permeability of valacyclovir in wild-type and PepT1 knockout mice. Drug Metab Dispos 41, 608-614.

Yang, B., Smith, D.E., 2017. In Silico Absorption Analysis of Valacyclovir in Wildtype and Pept1 Knockout Mice Following Oral Dose Escalation. Pharm Res 34, 2349-2361.

ACCEPTED MANUSCRIPT

ACCEPTED MANUSCRIPT

Table 1 Human Gastrointestinal Physiology

Compartment	ASF	pH	Transit Time (h)	Volume (mL)	Length (cm)	Radius (cm)	Hydrolase Expression	hPEPT1 Apical Expression (mg) [a]
Duodenum	2.872	6.00	0.25	35.56	13.08	1.47	0.802	16.28
Jejunum 1	2.867	6.20	0.91	129.5	54.06	1.38	0.803	87.84
Jejunum 2	2.831	6.40	0.73	104.7	54.06	1.24	0.808	73.93
Ileum 1	2.837	6.60	0.57	79.33	54.06	1.08	0.799	78.61
Ileum 2	2.772	6.90	0.41	60.62	54.06	0.94	0.816	63.41
Ileum 3	2.748	7.40	0.28	42.36	54.06	0.79	0.810	49.29
Caecum	0.018	6.40	3.77	40.87	12.30	3.25	0.831	0.0
Asc Colon	0.050	6.80	11.32	43.38	25.73	2.32	0.822	0.0

[a]: Reference Drozdzik et al., 2014; Groer et al., 2013

Table 2 Physicochemical and biopharmaceutical properties of acyclovir and valacyclovir used in the PBPK models

Input parameter	Acyclovir-IV	Acyclovir-Oral	Valacyclovir-Oral
Molecular weight	225.21 [a]	225.21 [a]	324.34 [a]
Log P	-1.56 [a]	-1.56 [a]	-0.85 [b]
pK_a	9.25 (Acid)	9.25 (Acid)	9.4 (Acid)
	2.27 (Base) [a]	2.27 (Base) [a]	7.4 (Base), 1.9 (Base) [c]
Dose (mg)	350	200	1000
Solubility (mg/mL @pH=7)	2.5 [a]	2.5 [a]	174 [c]
Blood/plasma conc. ratio	1 [d]	1 [d]	0.81 [b]
Use Exp Plasma Fup (%)	99 [d]	99 [d]	82.94 [b]
Passive Peff (cm/s×10^4)	0.19 [d]	0.19 [d]	0.19 [d]

ACCEPTED MANUSCRIPT

Vss (L)	32.905 [e]	27.138 [e]	31.725 [e]
Clearance	Liver：18 [f]	Liver：18 [f]	Liver：56.4 [f] Blood：13.00 [f]
Mean precipitation time (sec)	900 [e]	900 [e]	900 [e]
Diffusion coefficient, ($cm^2/s \times 10^5$)	0.97 [e]	0.97 [e]	0.74 [e]
Formulation option	IV: Infusion	IR: Tablet	IR: Tablet

[a]: Drugbank

[b]: Predicted by ADMET Predictor ™

[c]: Reference Bolger et al., 2009

[d]: Optimizing value based on ADMET Predictor ™ value

[e]: Default value in GastroPlus ™

[f]: Optimizing value based on observed plasma concentration-time curve

Table 3 Demographic information of PBPKmodel-1, PBPKmodel-2 and PBPKmodel-3.

Demographic parameter	**PBPKmodel-1 [a]**	**PBPK model-2 [b]** **PBPK model-3 [b]**
Species	Human	Human
Population	American	American
Gender	Female	Female
Health Status	Healthy	Healthy
Age (years)	36	35
Weight (Kg)	73	60
Height (cm)	162.72	162.74
BMI(kg/m^2)	29.1603	29.0398

[a]: Reference Soul-Lawton et al., 1995

[b]: Reference Neves et al., 2010

ACCEPTED MANUSCRIPT

Table 4 The predicted and observed values of C_{max} and $AUC_{0-\infty}$ after each dose administration of valaciclovir to normal volunteers

Dose (mg)	C_{max} (μg/mL)		$AUC_{0-\infty}$ (μg·h/mL)	
	Predicted	**Observed**	**Predicted**	**Observed**
250	1.61	2.24±0.28	3.61	5.59±1.04
500	2.76	3.45±1.10	7.19	10.66±1.70
1000	4.56	4.80±1.48	14.27	18.30±2.94
1500	6.17	6.19±1.63	21.28	24.04±5.80
2000	7.53	8.30±1.43	27.89	29.50±6.36

ACCEPTED MANUSCRIPT

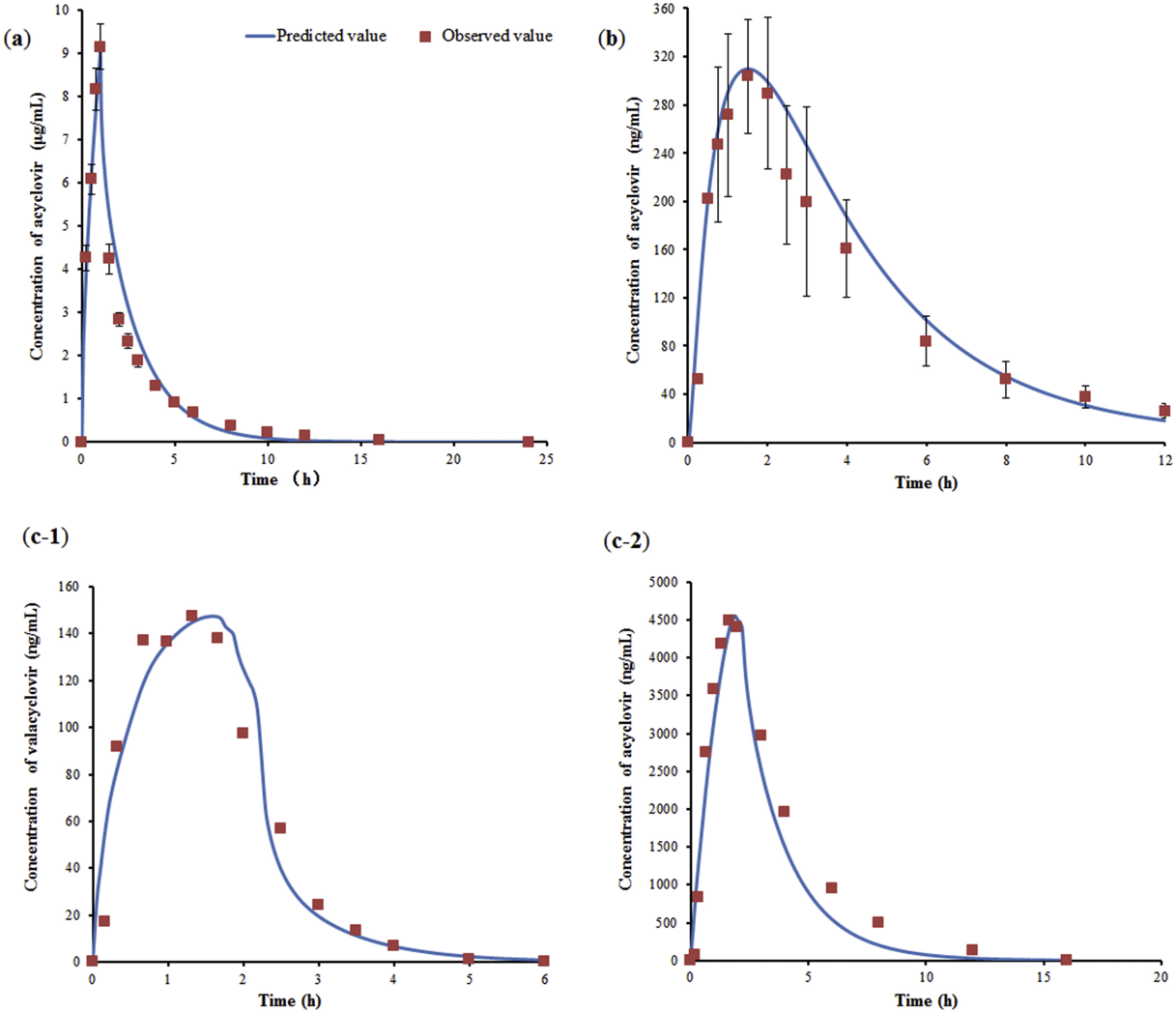

(a)
Predicted value
Observed value
Concentration of acyclovir (μg/mL)
Time (h)
(b)
Concentration of acyclovir (ng/mL)
Time (h)
(c-1)
Concentration of valacyclovir (ng/mL)
Time (h)
(c-2)
Concentration of acyclovir (ng/mL)
Time (h)

Figure 1

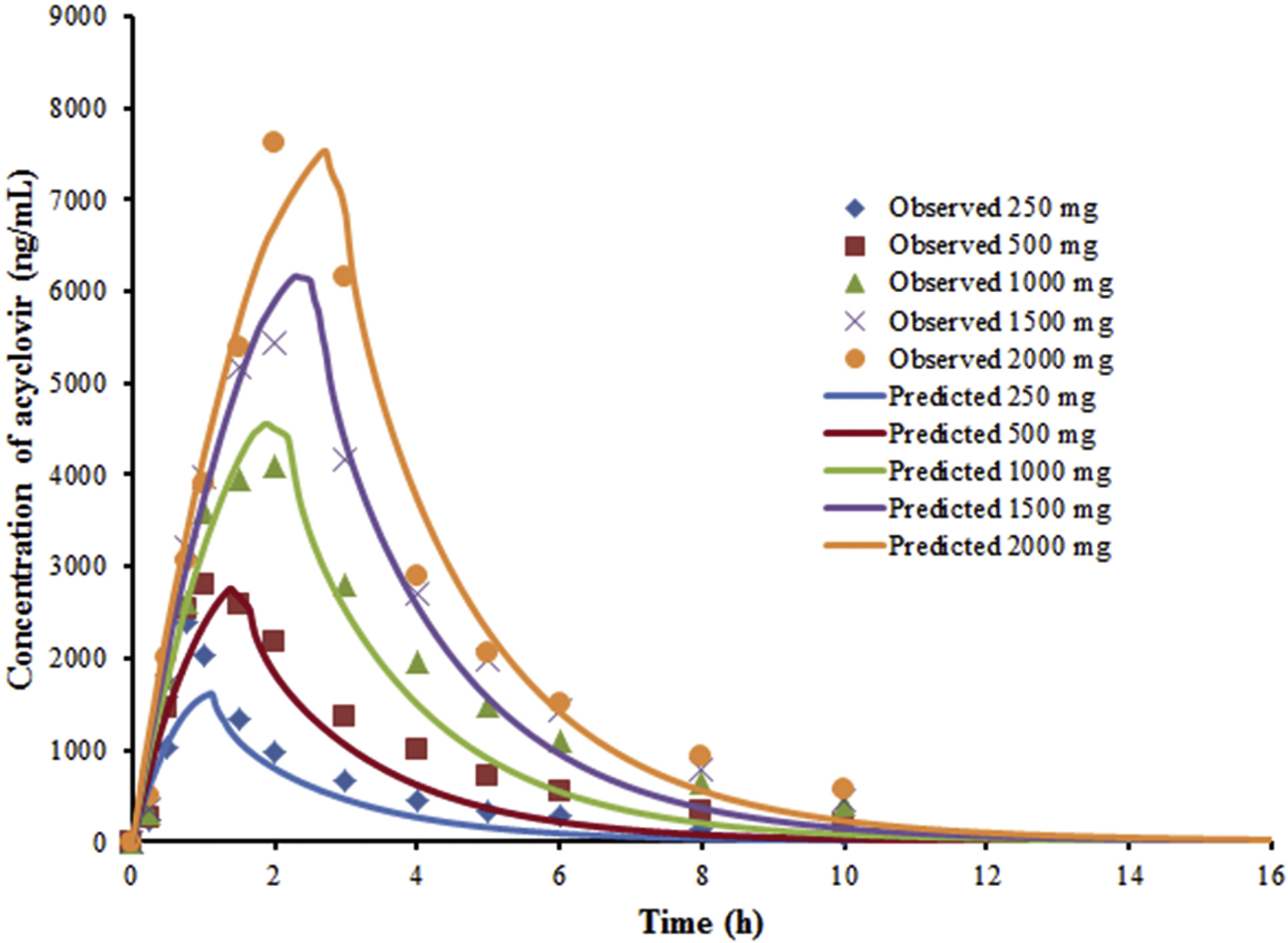

Concentration of acyclovir (ng/mL)
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
0
2
4
6
8
10
12
14
16
Time (h)
Observed 250 mg
Observed 500 mg
Observed 1000 mg
Observed 1500 mg
Observed 2000 mg
Predicted 250 mg
Predicted 500 mg
Predicted 1000 mg
Predicted 1500 mg
Predicted 2000 mg

Figure 2

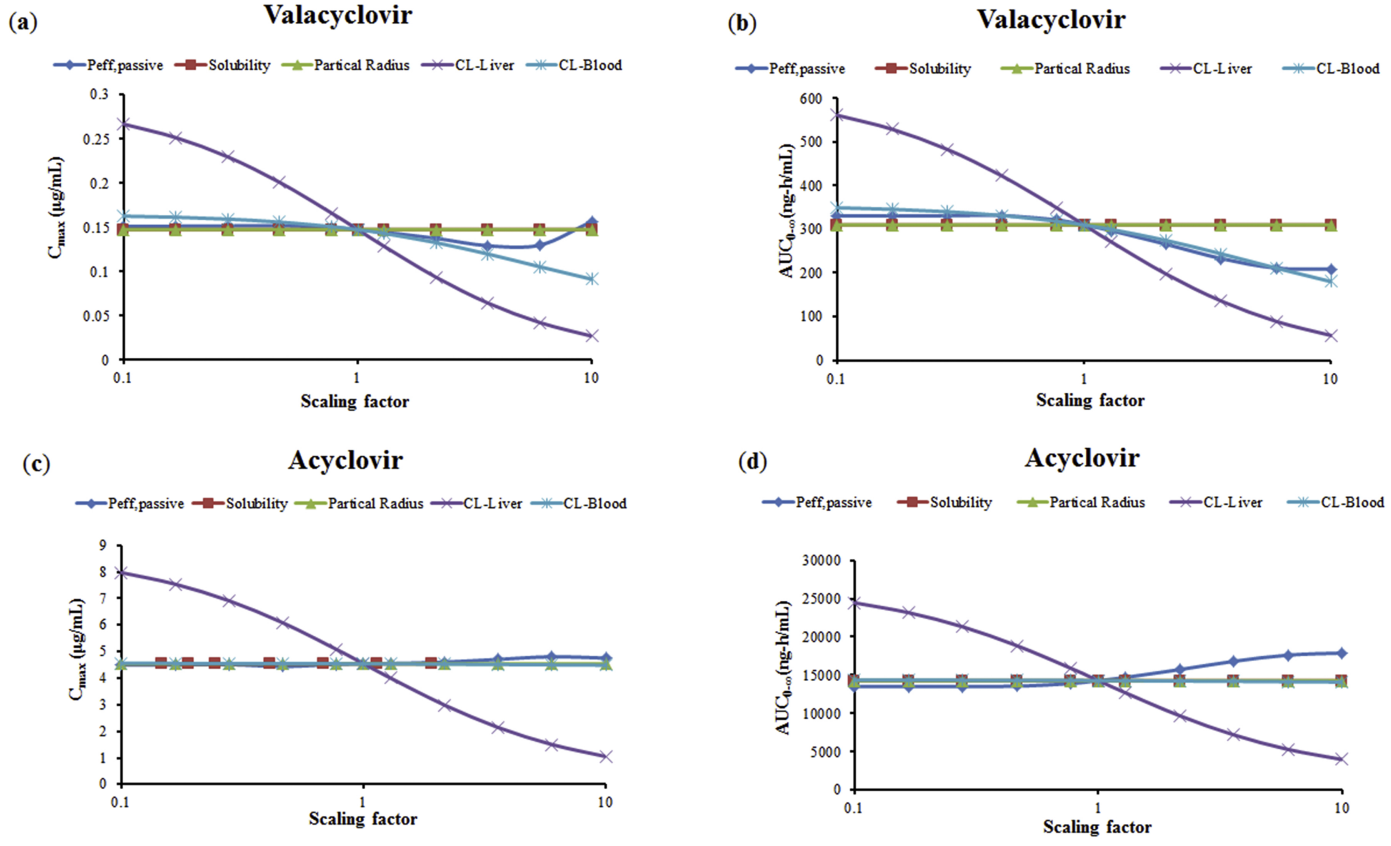

(a)
Valacyclovir
Peff,passive
Solubility
Partical Radius
CL-Liver
CL-Blood
Cmax (μg/mL)
Scaling factor
(b)
Valacyclovir
AUC0-∞ (ng-h/mL)
Scaling factor
(c)
Acyclovir
Cmax (μg/mL)
Scaling factor
(d)
Acyclovir
AUC0-∞ (ng-h/mL)
Scaling factor

Figure 3

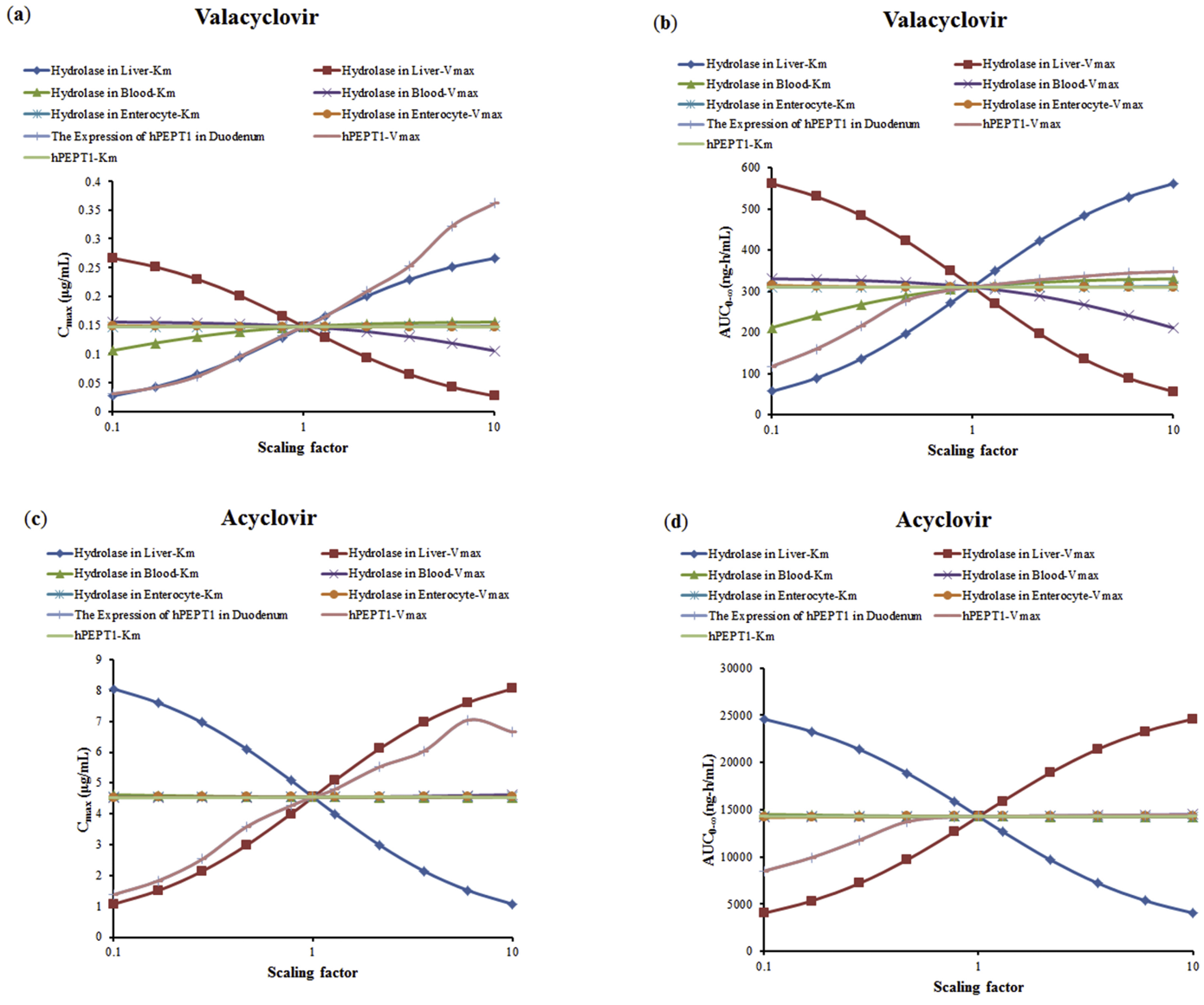

(a)
Valacyclovir
Hydrolase in Liver-Km
Hydrolase in Liver-Vmax
Hydrolase in Blood-Km
Hydrolase in Blood-Vmax
Hydrolase in Enterocyte-Km
Hydrolase in Enterocyte-Vmax
The Expression of hPEPT1 in Duodenum
hPEPT1-Vmax
hPEPT1-Km
Cmax (μg/mL)
Scaling factor
(b)
Valacyclovir
AUC0-∞ (ng-h/mL)
Scaling factor
(c)
Acyclovir
Cmax (μg/mL)
Scaling factor
(d)
Acyclovir
AUC0-∞ (ng-h/mL)
Scaling factor

Figure 4

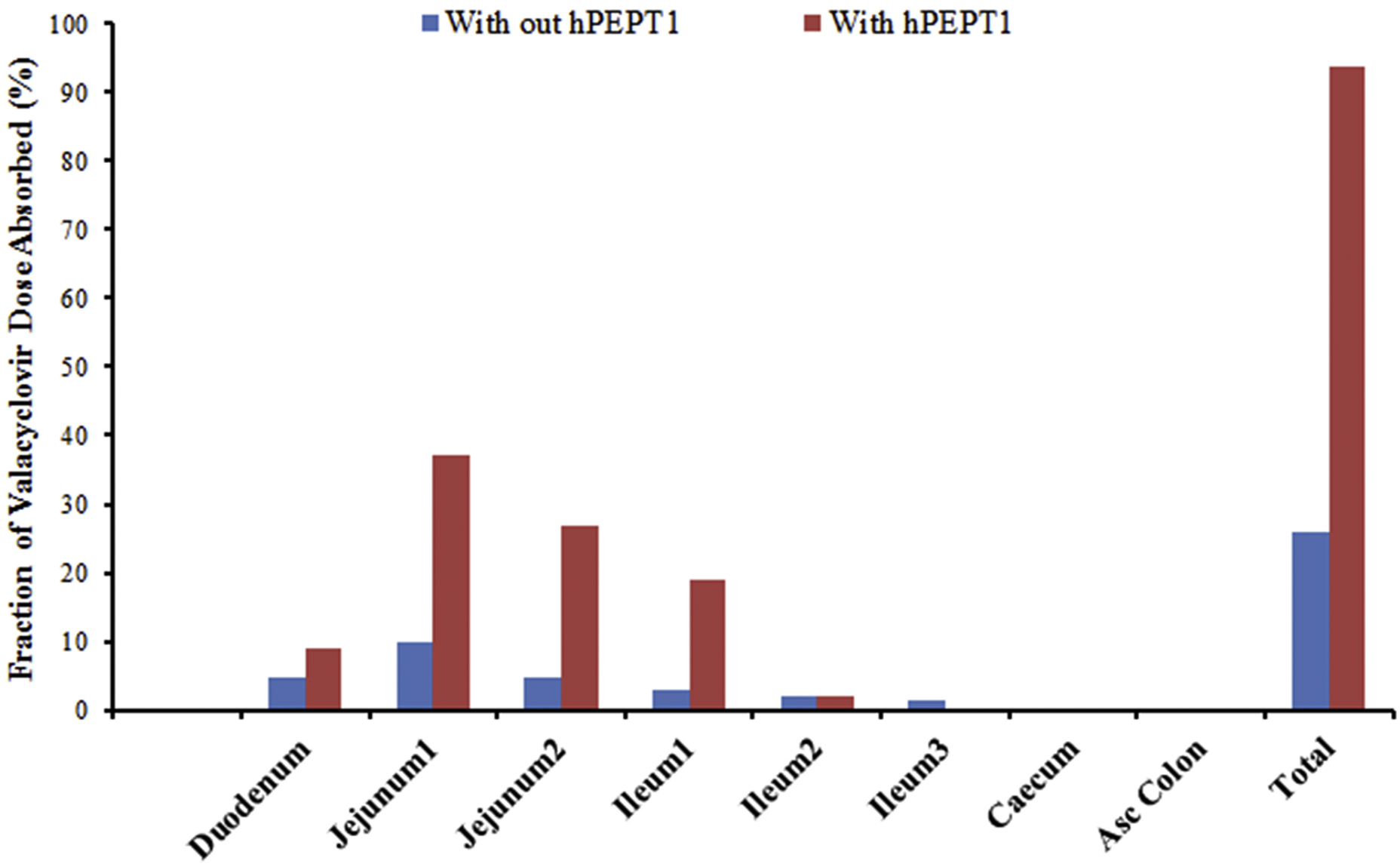
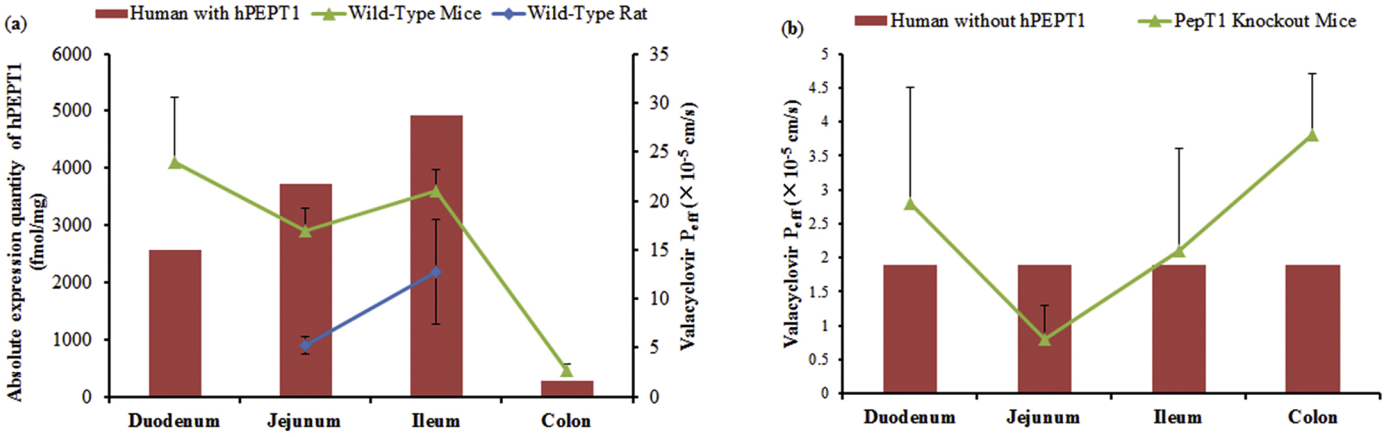

With out hPEPT1
With hPEPT1
Fraction of Valacyclovir Dose Absorbed (%)
100
90
80
70
60
50
40
30
20
10
0
Duodenum
Jejunum1
Jejunum2
Ileum1
Ileum2
Ileum3
Caecum
Asc Colon
Total

Figure 5



The effective permeability in wild type mice and PepT1 knockout mice duodenum, jejunum, ileum and colon were collected from literature (Yang and Smith, 2013)

Figure 6

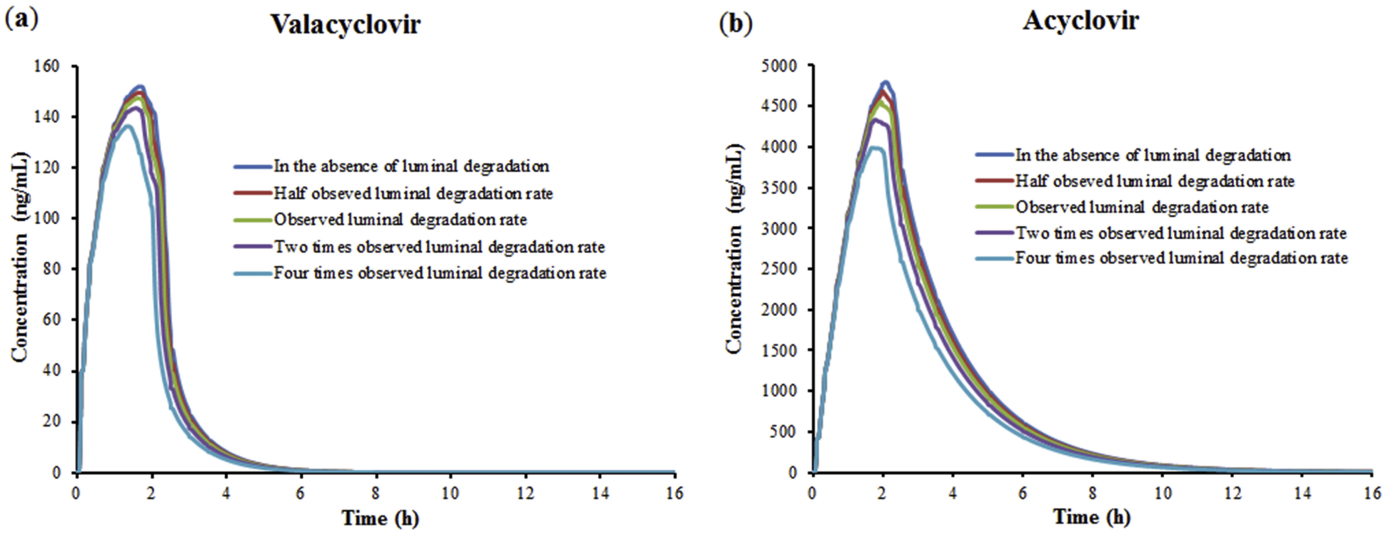

(a)
Valacyclovir
Concentration (ng/mL)
Time (h)
In the absence of luminal degradation
Half obsevved luminal degradation rate
Observed luminal degradation rate
Two times observed luminal degradation rate
Four times observed luminal degradation rate
(b)
Acyclovir
Concentration (ng/mL)
Time (h)
In the absence of luminal degradation
Half obsevved luminal degradation rate
Observed luminal degradation rate
Two times observed luminal degradation rate
Four times observed luminal degradation rate

Figure 7

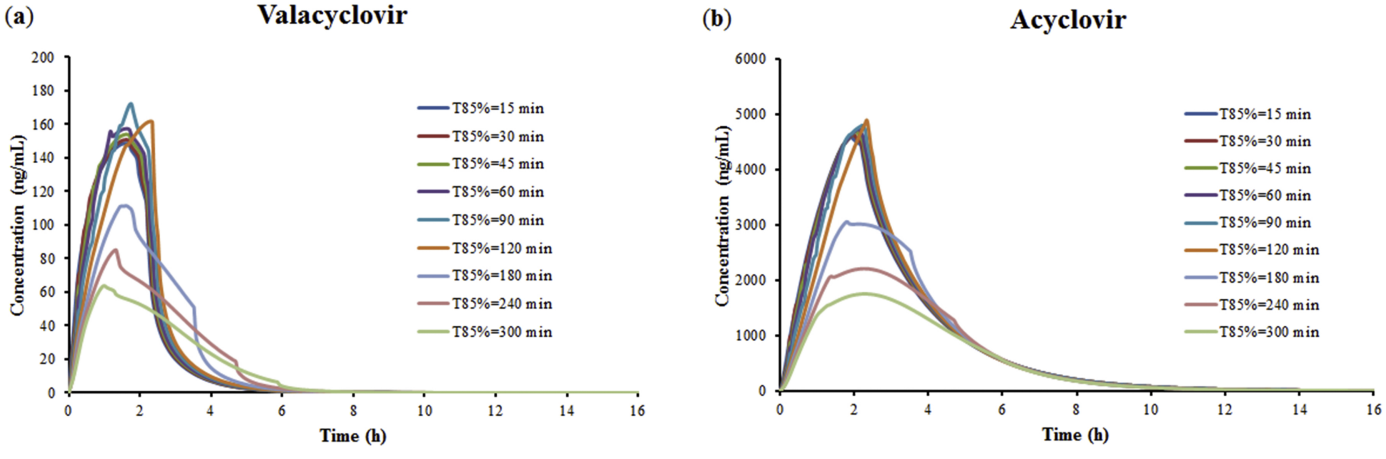

(a)
Valacyclovir
Concentration (ng/mL)
Time (h)
T85%=15 min
T85%=30 min
T85%=45 min
T85%=60 min
T85%=90 min
T85%=120 min
T85%=180 min
T85%=240 min
T85%=300 min
(b)
Acyclovir
Concentration (ng/mL)
Time (h)
T85%=15 min
T85%=30 min
T85%=45 min
T85%=60 min
T85%=90 min
T85%=120 min
T85%=180 min
T85%=240 min
T85%=300 min

Figure 8

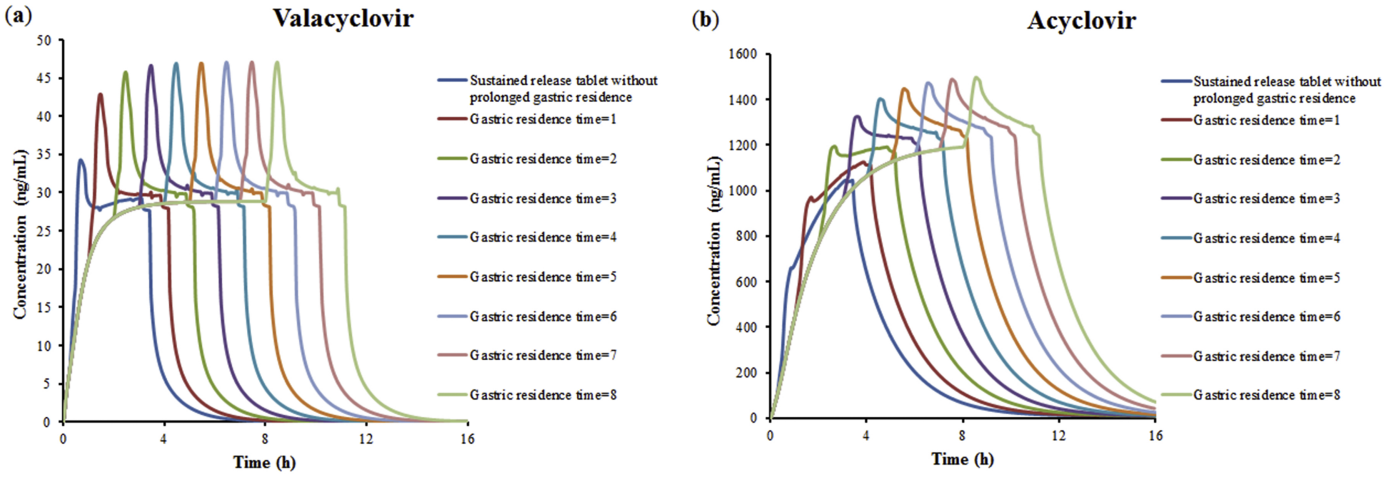

(a)
Valacyclovir
Concentration (ng/mL)
Time (h)
Sustained release tablet without prolonged gastric residence
Gastric residence time=1
Gastric residence time=2
Gastric residence time=3
Gastric residence time=4
Gastric residence time=5
Gastric residence time=6
Gastric residence time=7
Gastric residence time=8
(b)
Acyclovir
Concentration (ng/mL)
Time (h)
Sustained release tablet without prolonged gastric residence
Gastric residence time=1
Gastric residence time=2
Gastric residence time=3
Gastric residence time=4
Gastric residence time=5
Gastric residence time=6
Gastric residence time=7
Gastric residence time=8

Figure 9